Regression/Classification with Linear Models

Machine Learning

Nomenclature for supervised learning

$x = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$ features (aka: inputs, independent variables)

y output (aka: response, dependent variable)

$y \in \mathbb{N}$ classification

$y \in \mathbb{R}$ regression

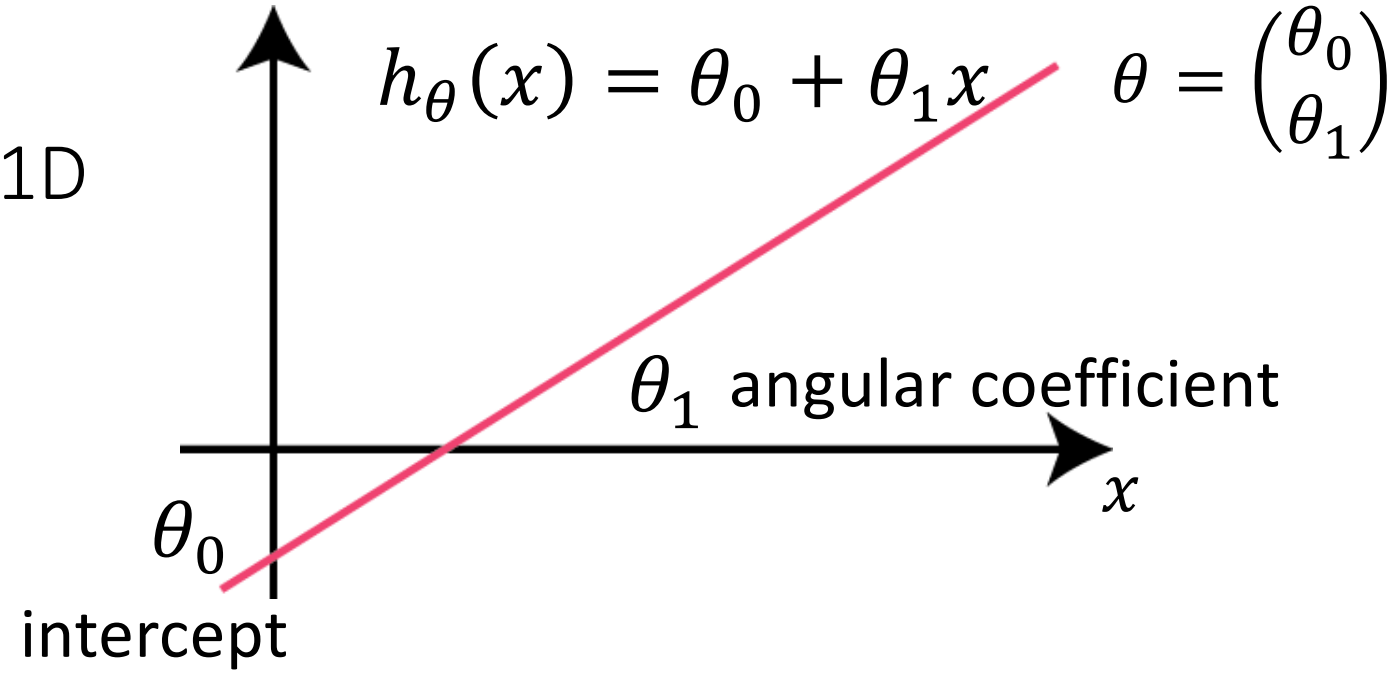
$h_\theta(x)$ hypothesis (prediction)

parameters of the algorithm

Linear Regression: the hypothesis is a linear function of the features

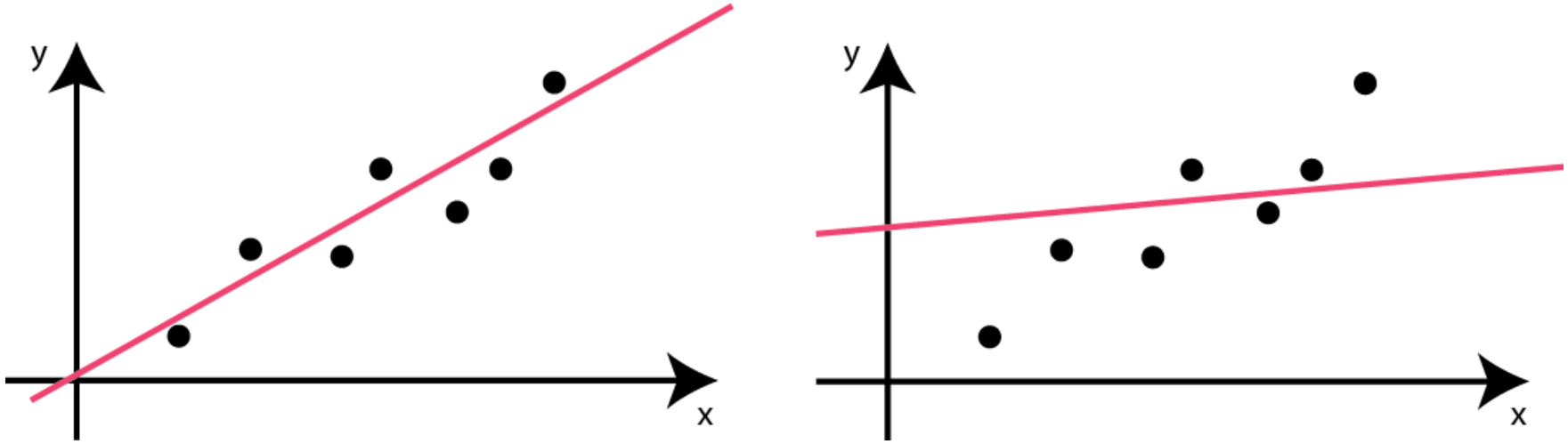
Linear Regression in 1D

$x \in \mathbb{R}$



$h_\theta(x) = \theta_0 + \theta_1 x$ How do we choose the parameters ?

Minimize the difference between the hypothesis and the output in the training set



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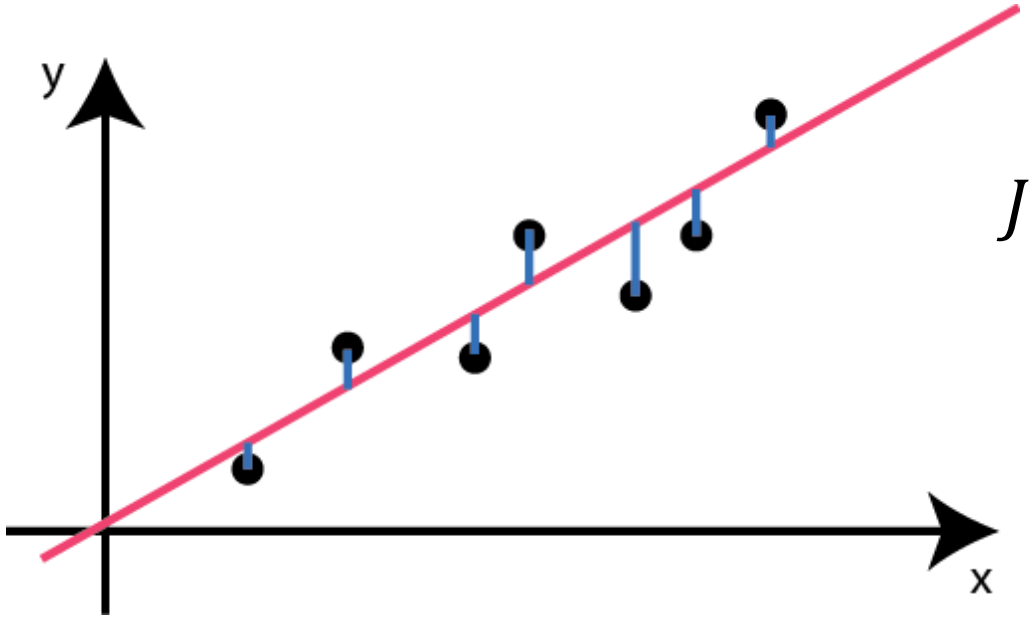


$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=0}^{m-1} \left(h_\theta\left(x^i\right) - y^i\right)^2$$

COST FUNCTION

Linear Regression with multiple features

$$h_\theta(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3 + \cdots + \theta_n x_n$$

$$h_\theta(x) = \theta_0 1 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3 + \cdots + \theta_n x_n$$

$$h_\theta(x) = \sum_{i=0}^{n} \theta_i x_i = \theta^T x$$

$$x_0 = 1$$

Linear Regression – Vectorial form

$$\Theta = \begin{pmatrix} \theta_0 \\ \vdots \\ \theta_n \end{pmatrix} \qquad x = \begin{pmatrix} x_0 \\ \vdots \\ x_n \end{pmatrix} \qquad h_\theta(x) = \sum_{i=0}^{n} \theta_i x_i = \Theta^T x$$

$$X = \begin{pmatrix} x_0^0 & \dots & x_n^0 \\ \vdots & \ddots & \vdots \\ x_0^{m-1} & \dots & x_n^{m-1} \end{pmatrix} = \begin{pmatrix} (x^1)^T \\ \vdots \\ (x^m)^T \end{pmatrix} \qquad X\Theta = \begin{pmatrix} h_\theta(x^0) \\ \vdots \\ h_\theta(x^{m-1}) \end{pmatrix} \qquad Y = \begin{pmatrix} y^0 \\ \vdots \\ y^{m-1} \end{pmatrix}$$

samples

features

$$J(\Theta) = \frac{1}{2m} \sum_{i=0}^{m-1} \left(h_\theta\left(x^i\right) - y^i \right)^2 \qquad J(\Theta) = \frac{1}{2m} (X\Theta - Y)^T (X\Theta - Y)$$

Gradient of the cost function – Vectorial form

$$\frac{\partial J}{\theta_j} = \frac{1}{2m}\sum_i \frac{\partial\left(h_\theta\left(x^i\right)-y^i\right)^2}{\partial\theta_j} = \frac{1}{m}\sum_i\left(h_\theta\left(x^i\right)-y^i\right)\frac{\partial h_\theta\left(x^i\right)}{\partial\theta_j}$$

$$= \frac{1}{m}\sum_i\left(h_\theta\left(x^i\right)-y^i\right)x_{ij}$$

$$\nabla_\theta J = \frac{1}{m}X^T(X\Theta - Y)$$

$$\Theta' = \Theta - \frac{\alpha}{m}X^T(X\Theta - Y)$$

Gradient Descent

Normal Equation

$$\nabla_\theta J = 0$$

$$X^T(X\Theta - Y) = 0$$

$$X^T X\Theta = X^T Y$$

$$\Theta = (X^T X)^{-1} X^T Y$$

- It's an exact solution
- It could be impossible to invert the matrix

Logistic Regression - Hypothesis

$y \in \{0,1\}$

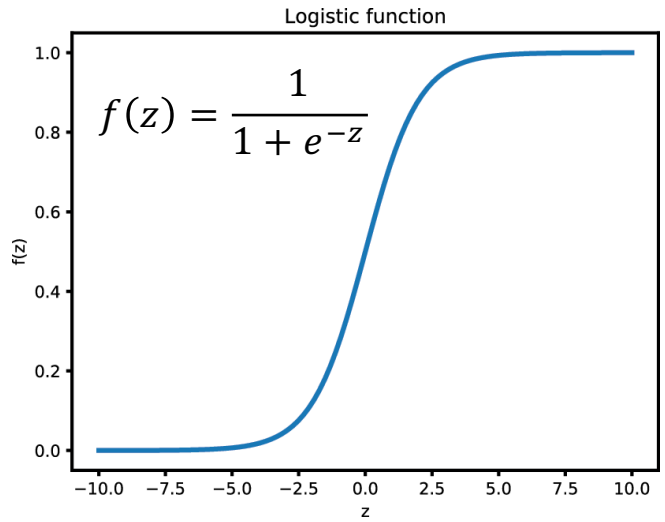
$h_\theta(x) = P(y = 1|x;\ \theta)$

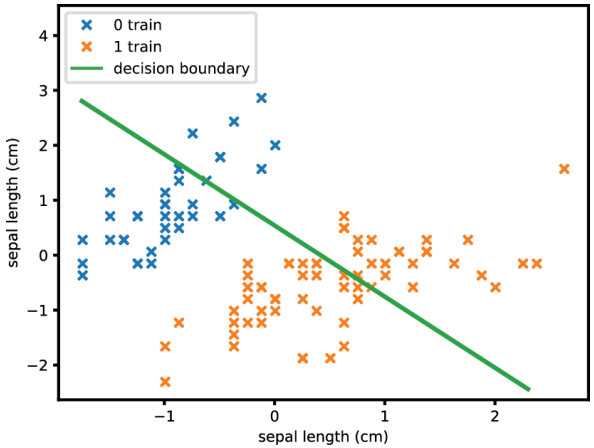
Probability that the sample with value x belongs to class 1, when using the parameters θ

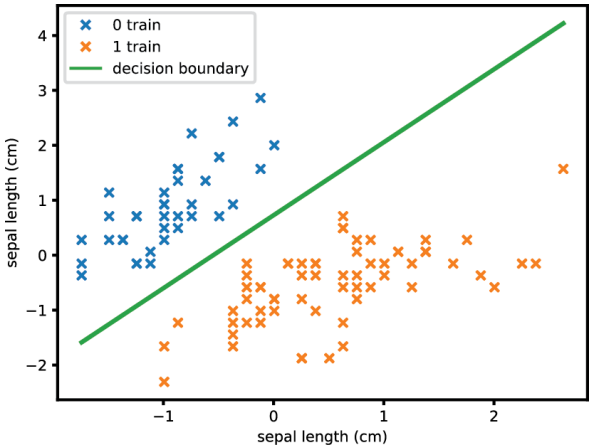
$$h_\theta(x) = f(\Theta^T x) = \frac{1}{1 + e^{-\Theta^T x}}$$

$$\Theta^T x = 0 \rightarrow f(\Theta^T x) = 0.5$$

Decision boundary




0 train
1 train
decision boundary
sepal length (cm)
sepal length (cm)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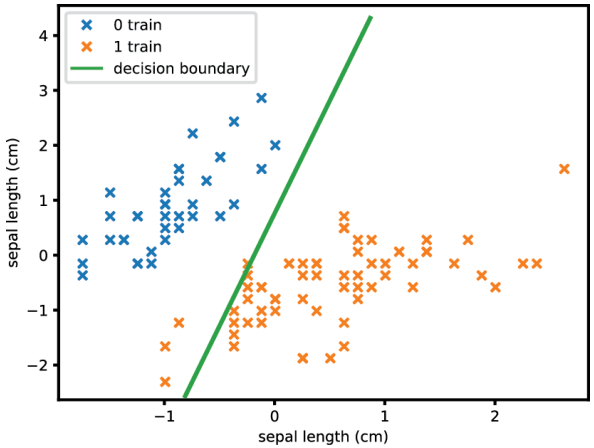
0 train
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Logistic Regression – Cost Function

$$J(\theta) = \frac{1}{m}\sum_i \frac{1}{2}\left(h_\theta\left(x^i\right) - y^i\right)^2 = \frac{1}{m}\sum_i cost\left(h_\theta\left(x^i\right), y^i\right)$$

cost function for linear regression

This function is not convex for logistic regression → hard to find the minimum

$$cost\left(h_\theta\left(x^i\right), y^i\right) = \begin{cases} -log\left(h_\theta\left(x^i\right)\right) \ if \ y^i = 1 \\ -log\left(1 - h_\theta\left(x^i\right)\right) \ if \ y^i = 0 \end{cases}$$

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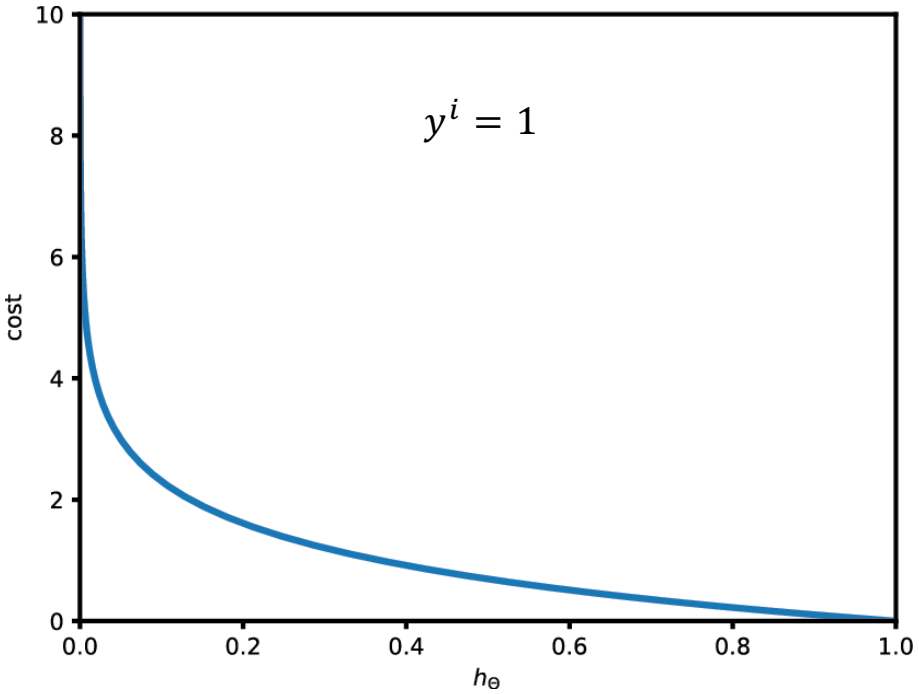
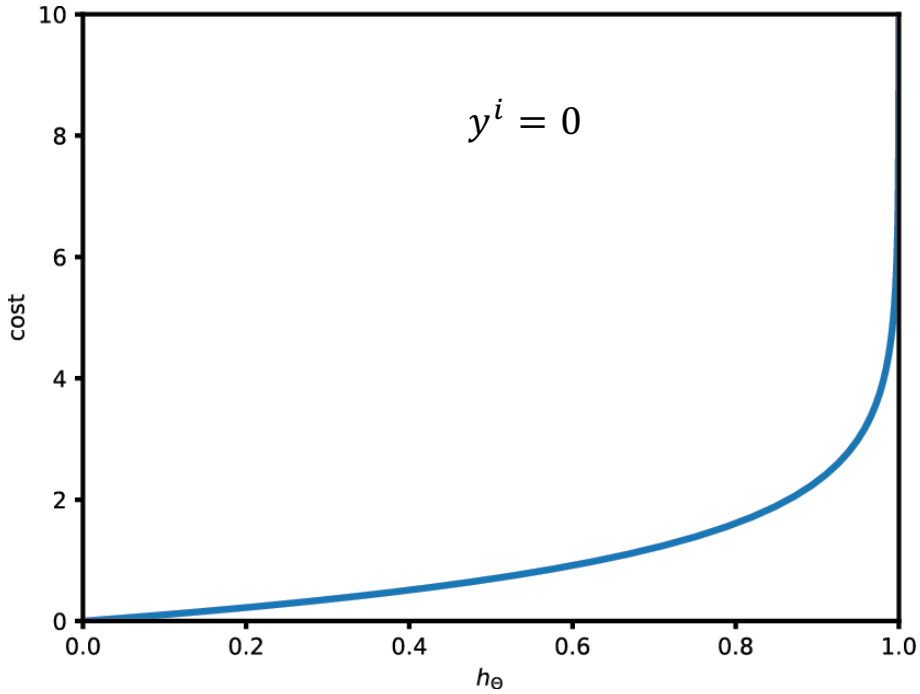
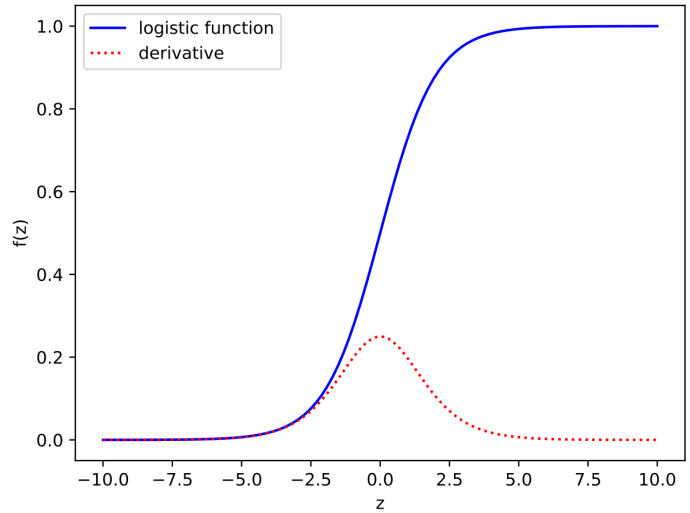




$$cost\left(h_\theta\left(x^i\right), y^i\right) = \begin{cases} -log\left(h_\theta\left(x^i\right)\right) \ if \ y^i = 1 \\ -log\left(1 - h_\theta\left(x^i\right)\right) \ if \ y^i = 0 \end{cases}$$

$$cost\left(h_\theta\left(x^i\right), y^i\right) = -y^i log\left(h_\theta\left(x^i\right)\right) - \left(1 - y^i\right) log\left(1 - h_\theta\left(x^i\right)\right)$$

$$J(\theta) = -\frac{1}{m}\sum_{i=1}^{m} y^i log\left(h_\theta\left(x^i\right)\right) + \left(1 - y^i\right) log\left(1 - h_\theta\left(x^i\right)\right)$$



$$f(z) = \frac{1}{1 + e^{-z}}$$

$$\frac{df(z)}{dz} = \frac{d}{dz}(1 + e^{-z})^{-1} = \frac{e^{-z}}{(1 + e^{-z})^2} = \frac{+1 - 1 + e^{-z}}{(1 + e^{-z})(1 + e^{-z})}$$

$$= \frac{1}{(1 + e^{-z})} \frac{1 + e^{-z} - 1}{(1 + e^{-z})} = f(z)\left(\frac{1 + e^{-z}}{1 + e^{-z}} - \frac{1}{1 + e^{-z}}\right)$$

$$= f(z)(1 - f(z))$$

$$J(\theta) = -\frac{1}{m}\sum_{i=1}^{m} y^i log\left(h_\theta\left(x^i\right)\right) + \left(1-y^i\right) log\left(1-h_\theta\left(x^i\right)\right)$$

$$\frac{\partial J}{\partial \theta_j} = -\frac{1}{m}\sum_{i=1}^{m} y^i \frac{1}{h}\frac{\partial h}{\partial z}\frac{\partial z}{\partial \theta_j} - \left(1-y^i\right)\frac{1}{h}\frac{\partial h}{\partial z}\frac{\partial z}{\partial \theta_j} = \qquad \boxed{z = \theta^T x^i}$$

$$-\frac{1}{m}\sum_{i=1}^{m} y^i \frac{1}{h} h(1-h)\frac{\partial z}{\partial \theta_j} - \left(1-y^i\right)\frac{1}{1-h} h(1-h)\frac{\partial z}{\partial \theta_j}$$

$$= -\frac{1}{m}\sum_{i=1}^{m} y^i\,(1-h)x_j^i\; - \left(1-y^i\right)hx_j^i = -\frac{1}{m}\sum_{i=1}^{m} y^i x_j^i - y^i hx_j^i\; - hx_j^i + y^i hx_j^i$$

$$= \frac{1}{m}\sum_{i=1}^{m}\left(h-y^i\right)x_j^i$$

Gradient of the cost function – Vectorial form

$$\frac{\partial J}{\theta_j} = \frac{1}{2m}\sum_i \frac{\partial\left(h_\theta\left(x^i\right)-y^i\right)^2}{\partial\theta_j} = \frac{1}{m}\sum_i\left(h_\theta\left(x^i\right)-y^i\right)\frac{\partial h_\theta\left(x^i\right)}{\partial\theta_j}$$

$$= \frac{1}{m}\sum_i\left(h_\theta\left(x^i\right)-y^i\right)x_j^i$$

$$\nabla_\theta J = \frac{1}{m}X^T\left(h_\theta(X)-Y\right)$$

$$\Theta' = \Theta - \frac{\alpha}{m}X^T\left(h_\theta(X)-Y\right)$$ Gradient Descent

Multi-class classification / One-Vs-All

- Train N models that classify each class versus the other classes
- Choose the class with highest probability

Multi-class classification / One-Vs-One

- Train N*(N-1)/2 models that classify each class versus each other class
- Choose the class that wins most duels

Softmax regression / Multinomial logistic regression

Compute a score for each class $s_k(x) = \theta_k^T x$

Each class has its own weights vector

Estimate the probability of class k using the Softmax function

$$p_k(x) = \frac{exp(s_k(x))}{\sum_j exp\left(s_j(x)\right)}$$

Find the parameters by minimizing the cross-entropy

$$J(\Theta) = \frac{1}{m} \sum_{i=0}^{m-1} \sum_{k=0}^{K-1} -y_k^i log\left(p_k\left(x^i\right)\right)$$

equal to 1 if sample i belongs to class k

Entropy

$$H(p) = -\sum_i p_i log_2 p_i$$

Average amount of information that you get from one sample drawn from a given probability distribution

Tomorrow can be sunny/rainy with equal probability
The entropy is -0.5*log2(0.5)-0.5*log2(0.5) = 1
If I know the weather, my uncertainty decreases by 1 bit of information

The weather is classified into 4 equally likely classes.
The entropy is -4*0.25*log2(0.25) = 2
If I know the weather my uncertainty decreases by 2 bits of information

In case that the four classes have probabilities:

- class 0 with prob. 1/2 → -log2(0.5) = 1 (is number of bits I need to encode an experiment with 2 possible outcomes)
- class 1 with prob. 1/4 → -log2(0.25) = 2 (is number of bits I need to encode an experiment with 4 possible outcomes)
- classes 2 and 3 with prob. 1/8 → -log2(0.125) = 3 (is number of bits I need to encode an experiment with 8 possible outcomes)

The entropy is the weighted average of these terms = 0.5 + 0.25*2 + 0.125*3 + 0.125*3 = 1.75 bits

Cross-entropy

predicted distribution

$$H(p,q) = -\sum_i p_i log_2 q_i = -\sum_i p_i log_2 \frac{p_i}{p_i} q_i$$

true distribution

$$H(p,q) = -\sum_i p_i log_2 p_i + \sum_i p_i log_2 \frac{p_i}{q_i} = H(p) + KL(p,q)$$

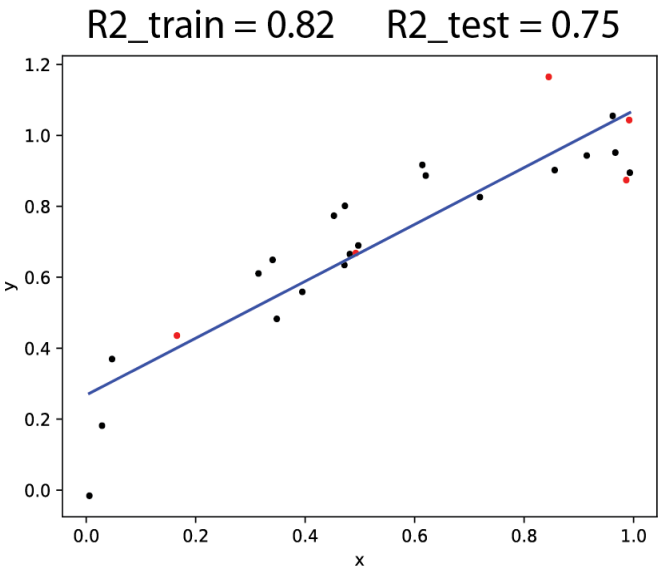
how many bits I need to encode the outcome using the wrong model q

entropy of the true distribution

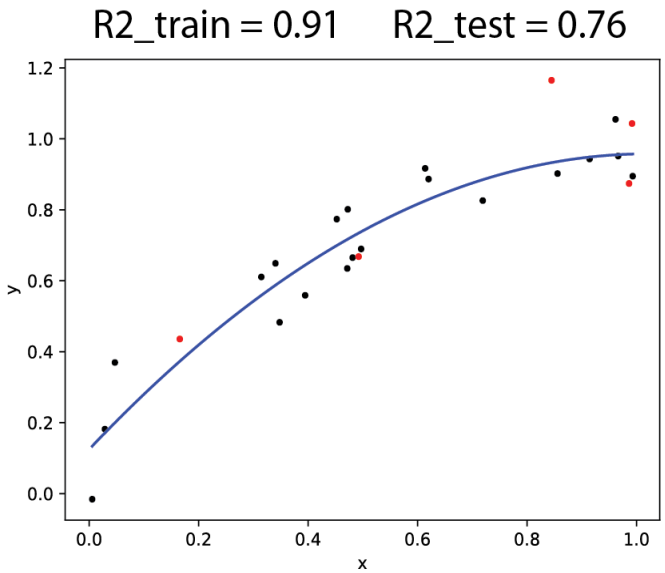
KL divergence: mismatch if p is approximated by q

So minimizing the cross-entropy corresponds to minimize the KL divergence

underfitting
aka: high bias

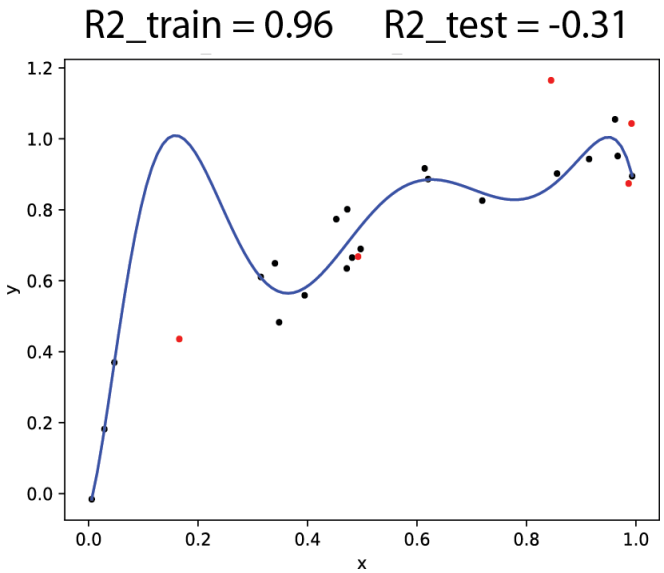


$h_\theta = \theta_0 + \theta_1 x$



$h_\theta = \theta_0 + \theta_1 x + \theta_2 x^2$

overfitting
aka: high variance



$h_\theta = \theta_0 + \theta_1 x + \theta_2 x^2 + \cdots + \theta_{10} x^{10}$

How to avoid overfitting

- Manually reduce the number of features (features' selection)
- Regularization = keep all the features but reduce the value of the parameters

$$J(\theta) = \frac{1}{2m}\left[\sum_{i=0}^{m-1}\left(h_\theta\left(x^i\right) - y^i\right)^2 + \lambda \sum_{j=1}^{n}\theta_j^2\right]$$

keep the model simple

reduce the distance from the true values in the training set

regularization parameter

The first parameter (intercept) is not usually included in this sum

$$J(\theta) = \frac{1}{2m}\left[\sum_{i=0}^{m-1}\left(h_\theta\left(x^i\right) - y^i\right)^2 + \lambda\sum_{j=1}^{n}\theta_j^2\right]$$

$$\frac{\partial J}{\partial \theta_j} = \frac{1}{m}\sum_i\left(h_\theta\left(x^i\right) - y^i\right)x_j^i \qquad j = 0$$

$$\frac{\partial J}{\partial \theta_j} = \frac{1}{m}\sum_i\left(h_\theta\left(x^i\right) - y^i\right)x_j^i + \frac{\lambda\theta_j}{m} \qquad j = 1,..,n$$

$$\theta_j = \theta_j - \alpha\left.\frac{\partial J}{\partial \theta_j}\right|_\theta = \theta_j\left(1 - \alpha\frac{\lambda}{m}\right) - \frac{\alpha}{m}\sum_{i=0}^{m-1}\left(h_\theta\left(x^i\right) - y^i\right)x^i$$

"shrink" the previous estimate of the parameter

$$\frac{\partial J}{\partial \theta_j} = \frac{1}{m}\sum_i \left(h_\theta\left(x^i\right) - y^i\right) x_j^i \qquad j = 0$$

$$\frac{\partial J}{\partial \theta_j} = \frac{1}{m}\sum_i \left(h_\theta\left(x^i\right) - y^i\right) x_j^i + \frac{\lambda \theta_j}{m} \qquad j = 1, .., n$$

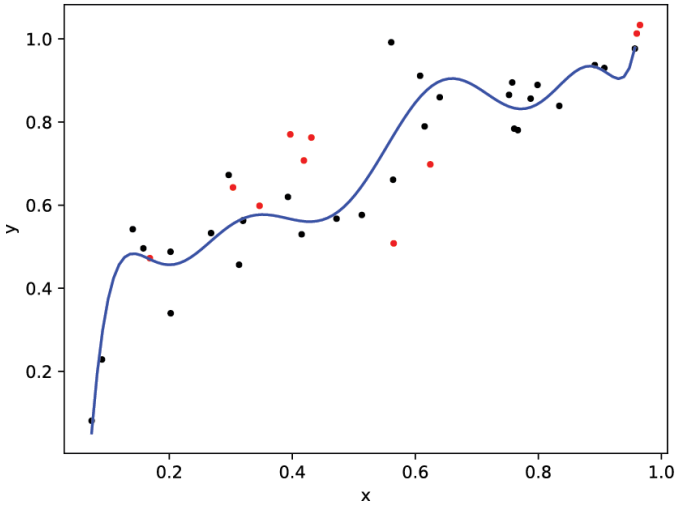
$$\nabla_\theta J = \frac{1}{m}\left[X^T(X\Theta - Y) + \lambda I_o \Theta\right] \qquad I_0 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \ddots & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$\Theta = \left(X^T X + \lambda I_o\right)^{-1} X^T Y$$

$$h_\theta = \theta_0 + \theta_1 x + \theta_2 x^2 + \cdots + \theta_{10}\, x^{10}$$

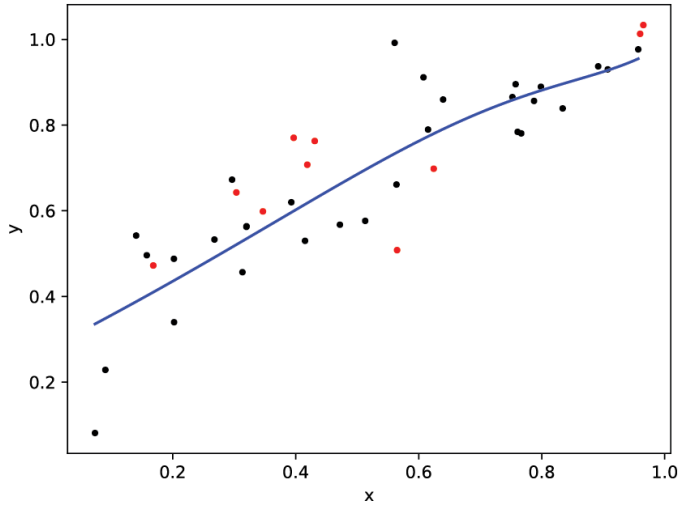
λ = 0

R2_train = 0.90 R2_test = 0.31



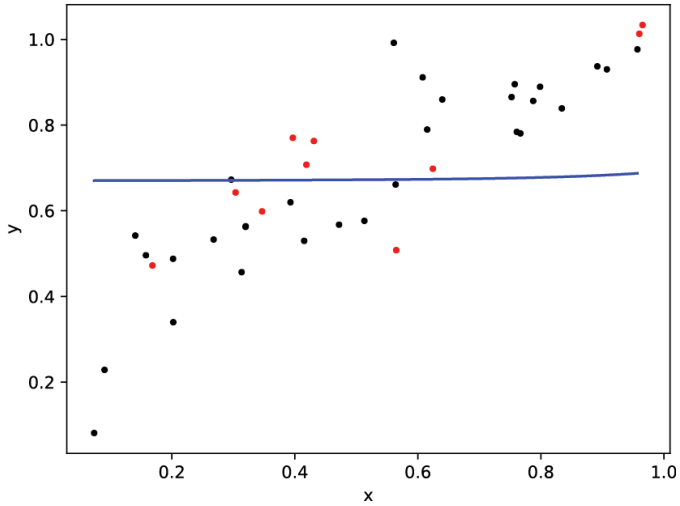
λ = 1.0

R2_train = 0.80 R2_test = 0.55



λ = 10000

R2_train = 0.02 R2_test = -0.01



Regularization methods

- Ridge (L2 regularization) $\frac{\lambda}{2m}\sum_{j=1}^{n}\theta_j^2$
- Lasso (L1 regularization) $\frac{\lambda}{2m}\sum_{j=1}^{n}|\theta_j|$

weights of non-relevant features are forced to be exactly zero

- Elastic nets
(L1 + L2 regularization)

$$\frac{\lambda}{2m}\left((1-r)\sum_{j=1}^{n}\theta_j^2 + r\sum_{j=1}^{n}|\theta_j|\right)$$

$$r \in [0,1]$$

Firth's Logistic Regression

Developed to deal with the problems that arise when using Logistic Regression for

- small datasets
- imbalanced datasets (the loss-function is dominated by the major class)
- complete separation of classes along one feature (the model tends to ±∞ depending on values of the independent variables, and consequently the parameters diverge)

$$J(\theta) = \frac{1}{2m}\sum_{i=0}^{m-1} cost(x^i, y^i) + \frac{1}{2} log|I(\theta)|$$

determinant of the Fisher's information matrix

Fisher Information Matrix

It measures how sensitive is the likelihood to changes in parameters

The idea is that if a small change in a parameter greatly change the likelihood, that parameter is highly informative

log-likelihood (or likelihood)

$$I(\theta) = -\mathbb{E}\left[\frac{\partial^2 l}{\partial\theta\partial\theta^T}\right]$$

$$I(\theta)_{ij} = -\mathbb{E}\left[\frac{\partial^2 l}{\partial\theta_i\partial\theta_j}\right]$$

Suppose the output is sampled from a Bernulli distribution with probability pi depending on the input features according to the logistic function

$$y_i \sim Bernulli(p_i) \qquad p_i(x) = \frac{1}{1 + e^{-\theta^T x}}$$

$$y_i \sim p_i^{y_i}(1 - p_i)^{1-y_i}$$

since samples are independent, the likelihood can be calculated as:

$$L(\theta) = \prod_i p_i^{y_i}(1 - p_i)^{1-y_i}$$

log-likelihood

$$l(\theta) = \sum_i y_i log p_i + (1 - y_i) log(1 - p_i)$$

$$\frac{\partial l}{\partial \theta_j} = \sum_i (y_i - p_i) x_{ij}$$

$$\nabla_\theta l = X^T (Y - P)$$

$$J(\theta) = -\frac{1}{m} \sum_{i=1}^{m} y^i log\left(h_\theta(x^i)\right) + (1 - y^i) log\left(1 - h_\theta(x^i)\right)$$

$$\frac{\partial J}{\partial \theta_j} = \frac{1}{m} \sum_{i=1}^{m} (h - y^i) x_j^i$$

$$X = \begin{pmatrix} x_0^0 & \dots & x_n^0 \\ \vdots & \ddots & \vdots \\ x_0^{m-1} & \dots & x_n^{m-1} \end{pmatrix}$$

$$Y = (y_1, \dots, y_m)^T$$

$$P = (p_1, \dots, p_m)^T$$

$$\frac{\partial^2 l}{\partial\theta_k \partial\theta_j} = \frac{\partial}{\partial\theta_k}\left(\sum_i (y_i - p_i)x_{ij}\right) = -\frac{\partial}{\partial\theta_k}\left(\sum_i x_{ij} p_i\right)$$

$$p_i(x) = \frac{1}{1 + e^{-\theta^T x}}$$

$$f(z) = \frac{1}{1 + e^{-z}}$$

$$\frac{df(z)}{dz} = f(z)(1 - f(z))$$

$$\frac{\partial^2 l}{\partial\theta_k \partial\theta_j} = -\sum_i x_{ij} \frac{\partial p_i}{\partial z} \frac{\partial z}{\partial\theta_k} = -\sum_i x_{ij} x_{ik} p_i (1 - p_i)$$

$$H(\theta) = -X^T W X$$

$$I(\theta) = X^T W X$$

$$W = \begin{pmatrix} p_1(1-p_1) & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & p_m(1-p_m) \end{pmatrix}$$

For a model with a single parameter, in the proximity of the maximum:

$$l(\theta+\delta) \approx l(\theta) + \frac{\partial l}{\partial \theta}\delta + \frac{1}{2}\frac{\partial^2 l}{\partial \theta^2}\delta$$

zero in the maximum of the likelihood

it's minus the information matrix

If the second derivative is high
→ a small change in the parameter causes a high change in the log-likelihood
→ high sensibility to the parameter

For a model with a n parameters:

$$l(\theta+\delta) \approx l(\theta) + \nabla_\theta l\delta + \frac{1}{2}\delta^T H_\theta \delta$$

$$l(\theta+\delta) \approx l(\theta) - \frac{1}{2}\delta^T I(\theta)\delta$$

In a similar way here, if the determinant of the information matrix is high, there's high curvature of the log-likelihood in the proximity of the maximum, which corresponds to high sensibility to the values of the parameters. Vice versa low-curvature means that there's low-dependency on the parameter values.

$$J(\theta) = \frac{1}{2m}\left[\sum_{i=0}^{m-1} cost(x^i, y^i)\right] + \frac{1}{2} log|I(\theta)|$$

So, the last term of the cost-function penalizes regions where the likelihood is flat, and consequently regions where the parameters are poorly defined.